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Review

Progress on agricultural residue-based microwave absorber: a review and prospects

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ABSTRACT

Microwave absorber comprising various agricultural residue-based microwave-absorbing materials is one of the interesting and challenging topics to modern-day scientists and engineers. In this work, we explicitly present an overview of the microwave absorption mechanism, different measurement techniques, various types of microwave absorber structure, and factors affecting the microwave absorption. We also discuss various stages of fabrication of lightweight and low-cost microwave absorber using different types of agricultural residues. This review provides an overall summary of the state-of-the-art progress on various agricultural residue-based microwave absorbers. Finally, some new prospects are also discussed from the existed theoretical and experimental results in this work, to propel their application in practice.

Introduction

Modern expeditious evolution in electronics and communication sectors and increasing use of electronic equipment and digital devices in every nook and cranny of our modern livelihood raise an alarming upsurge in the emission of electromagnetic (EM) radiation creating electromagnetic pollution. It leads to severe deterioration in electronic equipment, biological systems, information technology, and defense safety technologies. It has also a grievous effect on human health causing many diseases. As a result, this new kind of pollution has drawn serious worldwide attention and its subsidence can be achieved by either shielding mechanism or absorption mechanism. Compared to the shielding mechanism, the absorption mechanism has been received a great consideration to mitigate the EM pollution due to its effectiveness in dissipating microwave energy without any secondary pollution [1]. In recent times, microwave-absorbing materials (MAMs) are treated as a powerful alternative to eradicate EM interference. Good-quality MAM must have some general features as follows [2]:

- Reduce the reflection of EM waves at the free space from the absorber interface.
- Should have strong absorption of EM waves.

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- Wide bandwidth is desirable.
- Lightweight and low thickness are expected.

To face with these features and escalate the efficiency of MAM, various extensive techniques were discussed by authors [2–6]. These techniques can be classified into two prime categories: material based and geometry based. In the material-based technique, the electromagnetic parameters of the material, i.e., complex permittivity (ε) and permeability (μ), are tailored in a manner to increase electrical and magnetic losses of the material. In the geometry approach, the MA material is designed in a certain structure with particular geometry and dimensions which reduces the reflection and enhances confinement and absorption of the EM waves inside the structure. In recent years, there have been booming studies performed on carbon, metal, and metallic particles, conducting polymers, CNT, and grapheme [3–11]. The contemporary materials used in the developing of microwave absorber (MA) over the few decades are chemical based and ferrites based. The high fabrication cost, lack of bio-degradability, and toxic emission under high temperature compel us to quest for alternative materials. In general, carbon is used to fabricate MA as a primary element. Being a semiconductor material, carbon allows a small amount of charge to flow through it and has favorable electrical characteristics [1]. Agricultural residues having high carbon content have drawn great attention as a potential substitute material in the field of microwave communication with the benefit of less fabrication cost, low density, easy availability, and eco-friendly nature. The existing study on MAM using agricultural residues such as rice husk, sugarcane bagasse (SCB), coconut residues, and corn as well as absorption performance is extensively discussed in detail. The carbon content in various agricultural residues is enlisted in Table 1.

Table 1 Carbon content in different agricultural residues

Agricultural residues	Carbon content (%)	Refs
Rice husk	35.77	[12]
Coconut shell charcoal	83	[13]
Sugarcane bagasse	17.89	[14]
Banana leaves	43.5	[15]
Corn	79.11	[16]

Factors affecting the MA performance

Dielectric behavior

To develop a MA structure, there are two basic criteria as follows: (1) minimizing the reflection and (2) enhancing absorption bandwidth. The reflection can be minimized by wave impedance matching. Together, a strong absorption can be achieved by regulating the electromagnetic parameters of the materials like complex permittivity, loss tangent, and conductivity. The complex permittivity is a master cause of determining how the electromagnetic field interacts with the molecular structure of the material [1]. The complex permittivity can be expressed as follows [1, 17]:

$$\varepsilon = \varepsilon' - j\varepsilon'' \tag{1}$$

where ε' is known as the dielectric constant. It defines the material's ability to store and release electromagnetic energy. $\varepsilon''$is the loss factor or dissipative factor. It characterizes the material's ability to absorb or attenuate electromagnetic energy to create thermal energy. Loss tangent is another factor which points toward the efficiency of conversion of microwave energy into thermal energy within the dielectrics as given by [1]

$$\tan\delta = \frac{\varepsilon''}{\varepsilon'} \tag{2}$$

It is defined as the ratio of the energy dissipated to the energy stored in the dielectric material. The power loss, i.e., absorption of the incoming wave, can be explained in terms of heat generation due to the electrical component of the EM wave, and it can be further expounded by the following equation [1, 18].

$$P = \frac{1}{2}\sigma|E|^2 + \omega\varepsilon_0\varepsilon''|E|^2 \tag{3}$$

where P represents the power density in the material (W/m^3), σ is the electrical conductivity (S/m), $\omega = 2\pi f$ (Hz) where f is the frequency of the incident microwaves, ε'' is the dielectric loss factor, ε_0 is the permittivity of free space, and E represents the electric field strength (V/m).

Dielectric loss

When an electromagnetic wave impinged on a dielectric material, it dissipates electrical energy. The dissipated energy is then converted into heat energy. The energy produced is named as the dielectric loss. The mechanism of dielectric loss includes conductance loss and dielectric relaxation loss [19].

Conductivity and Conductance loss

The role of the conduction mechanism in wave absorption is associated with the grain and grain boundary. The relation between ac conductivity and frequency can be expressed by the following equation:

$$\sigma_{ac} = A(2\pi f)^2 + \sigma_{dc} \tag{4}$$

where A is the pre-exponential factor or characteristics factor, f is the frequency, σ_{ac} is typically assigned to the hopping conduction and ac conductivity, $s(\in (0,1))$ is the frequency component [20] with values in the range from 0 for ideal ionic-type crystals and to 1 for ideal Debye dielectric dipolar-type crystals [21], and σ_{dc} is the frequency-independent (dc conductivity) conductivity. The dc component is important at low frequencies, whereas the ac component dominates the high frequencies. The relation between the ac component of conductivity and the complex permittivity is given by [17]

$$\begin{aligned} \sigma_{ac} &= j\omega\varepsilon_0\varepsilon \\ \text{and } \sigma_{ac} &= \sigma' + j\sigma'' = j\omega\varepsilon_0(\varepsilon' - j\varepsilon'') \end{aligned} \tag{5}$$

The term σ' and σ'' are analogous to ε' and ε''. Therefore, the conductivity (σ') and conductance loss (σ'') can be expressed in terms of ε' and ε'' as follows

$$\begin{aligned} \sigma' &= \omega\varepsilon_0\varepsilon'' = 2\pi f\varepsilon_0\varepsilon'' = 2\pi f\varepsilon_0\varepsilon' \tan\delta \\ \text{and } \sigma'' &= \omega\varepsilon_0\varepsilon' = 2\pi f\varepsilon_0\varepsilon' \end{aligned} \tag{6}$$

The above equation implies that the variation in conductivity at low frequency is attributed to the polarization effect. In this polarization mechanism, the charged particles move back and forth through the material under the influence of the electric field, generating an electric current. These induced currents lead to heat generation in the materials due to electrical resistance caused by the collisions of charged particles with neighboring charged particles. In the intermediate frequency region, conductivity is almost found to be frequency independent and is equal to dc conductivity. In the high-frequency region, the conductivity increases with the frequency. In the high-frequency region, the increase in conductivity can be explained by the intensified hopping of charge carrier phenomenon [21]. Equation (6) implies that the conductance loss is higher at high frequencies.

Dielectric relaxation loss

When an electromagnetic field is applied across a dielectric material, an internal electric field is produced in the material. This filed induces the translational motion of electrons or ions and also a force to rotate the dipoles to align with the internal electric field [1]. Then the material is said to be polarized. This polarization mechanism is known as dipolar polarization. The inherent material properties like inertial, elastic, and frictional forces confine this induced motion as well as the realignment of dipoles [1]. Consequently, the change of polarization is slower than that of the electric field at increasing frequency and the dielectric relaxation loss is generated. It causes energy loss which is the cause of heat dissipation. The dielectric relaxation loss tangent can be evaluated using the Debye equation [22]:

$$\tan\delta_{rel} = \frac{\varepsilon''(\omega)}{\varepsilon'(\omega)} = \frac{(\varepsilon_s - \varepsilon_\infty)}{\varepsilon_s + \varepsilon_\infty(\omega\tau)^2} \tag{7}$$

where ε_s, ε_∞, and τ represent the permittivity at a frequency approaching zero and infinity, and relaxation time, respectively.

Size of the particles

In microwave absorption performance of a material, a factor which plays a cabalistic role is the size of the particles. If the particle size is small, it leads to a higher surface area, more surface atoms in the materials. It causes multiple reflections which in turn provide larger dielectric constant or/and dielectric loss, conductance loss, and lead to the interfacial polarization. Yielding of together is the cause of increasing microwave absorption performance [23].

Thickness

To obtain minimum reflection of the microwave signal, the thickness of the MA should be approximately

a quarter of the propagating wavelength multiplied by an odd number as described by the following equation [24]:

$$t_m = n\frac{\lambda_m}{4} \tag{8}$$

where n = (1, 3, 5, 7…), and n = 1 corresponds to the first dip at low frequency. The propagating wavelength (λ_m) is given by

$$\lambda_m = \frac{\lambda_0}{\sqrt{|\varepsilon||\mu|}} = \frac{C}{f\sqrt{|\varepsilon||\mu|}} \tag{9}$$

where ε and μ are the complex permittivity and permeability at frequency f and λ_0 is the free space wavelength of the incident microwaves. As the samples are dielectric (assumed $\mu = 1$), the above equation can be expressed as follows:

$$t_m = n\frac{C}{4f\sqrt{|\varepsilon|}} \tag{10}$$

If the calculated t_m agrees with the experimental matching thickness t_m, it suggests that MA performance agrees well with the quarter-wavelength matching model.

Types of absorbing materials

Two types of microwave absorbers are usually found which are impedance-matching absorber and resonant absorbers. A good-quality MA must have the following characteristics mentioned below:

Impedance matching

The traveling wave incident on the surface of an absorber interface will envisage some reflection which is proportional to the progressive transition in impedance between an incident and transmitting media. The impedance can be matched between the incident and absorbing media by using the graded dielectric absorber such as pyramidal, tapered loading, and matching absorber.

Pyramidal absorber In the pyramidal absorbers, a gradual transition occurs in impedance from air to that of the absorber. The height and periodicity of the pyramids rely on the order of one wavelength. These absorbers provide high absorption performance and wide bandwidth. Different types of pyramidal absorbers such as the wedge, truncated, and the oblique wedge were reported [25], and twisted pyramidal absorber, hollow pyramidal absorber, and wedge absorber were introduced in earlier [26].

The limitations of these absorbers are their thickness and tendency to fragile. They are usually used in the fabrication of anechoic chambers.

Tapered loading absorber Tapered loading absorbers are mostly in the form of a slab composed of low loss material and lossy material composition. The lossy material is uniformly and homogeneously composed and parallel to the surface with a gradient perpendicular [27].

The benefit of these absorbers over the pyramidal absorbers is their thinness. The limitation is that they do not exhibit excellent absorption performance.

Matching layer absorber In this absorber, a transition absorbing layer is used between the incident and absorbing media to lessen the thickness which is required for the gradual transition materials. The transition layer has the impedance values (i.e., the absorber and incident interface) to be matched. This matching takes place when the thickness of the matching layer is one-quarter of a wavelength of the incident wave [28, 29] and

$$Z_2 = \sqrt{Z_1 Z_3} \tag{11}$$

where Z_1, Z_2, and Z_3 are the impedance of the incident media, transition layer, and absorbing layer, respectively.

Resonant absorber

In this type of absorber, impedance matching between an incident and absorbing layer occurs merely and the thickness is less so that all the incident power is not absorbed. This type of layout causes reflection and transmission at the first layer. The reflected wave passes through a phase reversal of π. The transmitted wave propagates through an absorbing layer and is reflected from a metal backing. This second reflection causes a phase reversal of π before the wave returns to the incident medium. Different types of resonant absorbers [30–32] are discussed below.

Dallenbach absorber In the Dallenbach layer absorber [30], the lossy homogeneous dielectric layer of quarter wavelength is placed on a conductive backplate.

The thickness and permittivity of the layer are adjusted so that the reflection can be lessened for the desired wavelength. Two or more layers can be used to enhance the absorption bandwidth of this type of MA structure. Unfortunately, it suffers from narrow absorption bandwidth and low angle ranges.

Salisbury screen In the Salisbury screen absorber [31], a resistive sheet is placed at an odd multiple of one-quarter of a wavelength ($\lambda/4$) over the conductive metal backing separated by an air gap. The air gap can be replaced by a material with higher permittivity. Perfect reflection occurs when the gap is $\lambda/2$. The effective impedance of this MA structure is sheet resistance. The good impedance matching occurs when sheet resistance is 377 Ω. The limitation of this design is narrow absorption bandwidth and bulkiness.

Jauman absorber The Jauman absorber [32] uses the resistance sheets stacked over each other at a distance of $\lambda/4$. This structure was designed to obtain a wider absorption band compared to Salisbury screen absorber. However, the overall thickness of the absorber becomes wider than that required and the MA structure is quite complex in design.

Absorption mechanism

Absorption in terms of return loss

The components of the incident power (P_i) inside the material are reflected (P_r), absorbed (P_a) and transmitted power (P_t), respectively, which are expressed as [1]

$$P_i = P_r + P_a + P_t \tag{12}$$

The return loss can be expressed in terms of reflected power and incident power as given below

$$RL(dB) = 20\log\frac{P_r}{P} \tag{13}$$

Equation (14) shows the absorption profile AR(ω) of the material [1, 33]

$$\mathrm{AR}(\omega) = 1 - \mathrm{R} - \mathrm{T} = 1 - |\mathrm{S}_{11}|^2 - |\mathrm{S}_{21}|^2 \tag{14}$$

where R and T are the reflection signal and transmission signal, respectively, S_{11} and S_{21} are the reflection coefficient and the transmission coefficient, respectively.

Attenuation constant and skin depth

When the transverse EM wave is propagating through a material, exponential decay is observed with the distance of x by a factor $e^{-\alpha x}$ where attenuation constant (α) is given by [1, 34]

$$\alpha = A \times \frac{\omega}{C} \times \sqrt{\frac{\mu'\varepsilon'}{2}} \times \sqrt{\left[\sqrt{1+(tan\delta)^2} - 1\right]} \tag{15}$$

where $\omega = 2\pi f$ and f is the frequency of microwave; C is the speed of light in vacuum, μ' is the relative permeability of the material ($\mu' = 1$), and ε' and tanδ are the electromagnetic parameters of the material. A is the adjustment factor (A = 8.68/100 dB.Np^{-1}) to convert α in dB/cm. Equation (15) shows that the attenuation of microwave power for a material corresponds to ε' and tanδ.

The penetration depth, D_p, quantifies the depth, while the incident EM power penetrates until a certain attenuation is acquired. The penetration depth indicates the depth at which the power density has distorted to $1/e$ ($\sim$ 37%) of its initial value. Materials with a bigger imaginary part of complex permittivity exert high microwave energy absorption, and thus the incoming power decreases exponentially from the surface to the core region. For the dielectric materials, penetration depth, i.e., depth of the heated region, is given by the following equation [35]

$$D_p = \frac{\lambda_0\sqrt{\varepsilon'}}{2\pi\varepsilon''} \tag{16}$$

Perspective to minimize the reflection

Single layer

The single-layer MA structure is formed with a lossy absorbing layer and a perfect electrical conductor (PEC) as a reflective layer as shown in Fig. 1a. A percentage of the incident wave reflected at air from the surface of the lossy absorbing layer to an extent due to impedance mismatching. It leads to poor microwave absorption performance. The reflection loss (RL) in terms of microwave absorption performance is calculated by the transmission line theory [36]

$$Z_{in} = Z_0\sqrt{\mu/\varepsilon}tanh\left[j\left(\frac{2\pi fd}{C}\right)\sqrt{\varepsilon\mu}\right]$$

$$= Z_0\sqrt{1/\varepsilon}tanh\left[j\left(\frac{2\pi fd}{C}\right)\sqrt{\varepsilon}\right] \quad (17)$$

$$RL(dB) = 20log\left|\frac{(Z_{in} - Z_0)}{(Z_{in} + Z_0)}\right| \quad (18)$$

where ε is the complex permittivity, μ (assumed equal to 1 for dielectric material) is complex permeability, C is the velocity of light, f is the frequency of microwave, d is the thickness of the absorber, Z_{in} is the input impedance of the absorber, and Z_0 is the impedance of the free space.

Multilayer

The multilayer MA structure is used to overcome the limitation that exists in single-layer MA. This MA structure has the feature to match the wave impedance, increase the microwave absorption performance, and broaden the absorption bandwidth. The double-layer comprises impedance matching layer, electromagnetic wave loss layer, and reflective layer, i.e., PEC as shown in Fig. 1b. The first layer, having low dielectric permittivity as compared to the second layer, can transmit the EM wave with minimal reflection and absorption. Thereafter, the transmitted wave is propagated through the second layer composed of high dielectric loss materials. Absorption, reflection, transmission, and scattering of the incoming wave occur in this layer. The transmitted wave now goes to the reflected layer. The role of the third layer is to make the transmission wave back to the second layer, and as a result, the maximum absorption is taken place in the second layer. The input impedance of a plane wave incident normal to the surface of an absorber backed by a PEC can be expressed as [37]

$$Z_{in} = \eta_i\frac{Z_{i-1} + \eta_i tanh(\gamma_i d_i)}{\eta_i + Z_{i-1}\tanh(\gamma_i d_i)} \quad (19)$$

where η_i is the intrinsic impedance of the ith layer and is calculated from the equation given below

$$\eta = \sqrt{\frac{j\omega\mu}{\sigma_s + j\omega\varepsilon}} \quad (20)$$

For free space $\sigma_s = 0$, $\mu = \mu_0 = 4\pi \times 10^{-7}N/A^2$, and $\varepsilon = \varepsilon_0 = 8.854 \times 10^{-12}F/m$. Therefore, $\eta = \eta_0 \approx 377\Omega$.

$$\gamma_i = j(2\pi f/C)\sqrt{\mu_i\varepsilon_i} = j(2\pi f/C)\sqrt{\varepsilon_i} \quad (21)$$

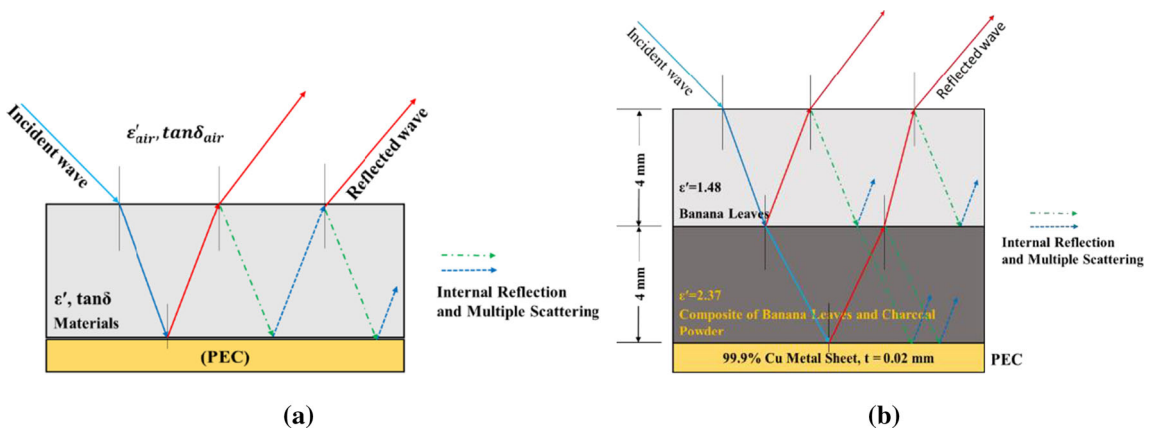
where $\mu_{i=1,2,...} = 1$(for dielectric material), and ε_i is the complex permittivity of the ith layer. The reflection loss (in dB) is calculated using Eq. (18).

Absorption performance

The transverse EM waves when propagating through the medium, exponential decay of the amplitude of the wave is experienced for the distance of x by a factor $e^{-\alpha x}$ ($E = E_m \times e^{-\alpha x}$), where α is the attenuation constant of the material and is a function of loss tangent as depicted in Eq. 15. The absorption of the incoming wave due to the influence of various parameters is expressed as [1, 38]

$$A = \frac{1}{2}\sigma|E|^2 + \frac{1}{2}\omega\varepsilon_0\varepsilon_r|E|^2 + \frac{1}{2}\omega\mu_0\mu_r|H|^2 \quad (22)$$

where E and H are the electric and magnetic field strength of incident wave, respectively, σ is the



(a) **(b)**

Figure 1 Distribution of the incident microwave in a material, **a** single-layer MA, and **b** double-layer MA.

conductivity of the material, ε_0 and μ_0 are permittivity and permeability of the free space, respectively, ε_r and μ_r are the relative permittivity and permeability of the material.

In the context of dielectric material, (22) can be rewritten as [1]

$$A = \frac{1}{2}\sigma|E|^2 + \frac{1}{2}\omega\varepsilon_0\varepsilon_r|E|^2 \quad (23)$$

Therefore, the absorption of the incident wave depends upon the attenuation constant of the material which is a function of loss tangent as well as the imaginary part of permittivity. It can be also elicited from (23) that the bigger the imaginary part of complex permittivity, the better the wave absorption effect [22].

Measurement techniques

Network analyzer instruments are commonly used to measure the dielectric properties and return loss profile in terms of absorption performance. There are two types of network analyzers such as scalar network analyzer (SNA) and vector network analyzer (VNA). The SNA measures signal amplitude only, and this is the cause not to use it in measuring complex signals. This limitation was excelled by VNA by measuring the amplitude of the signal along with various phases. The two ports correspond to the VNA point toward the reflected wave and transmitted wave in terms of S-parameters as shown in Fig. 2.

Different standard measurement methods are adopted to measure the complex permittivity and absorption performance of agricultural residue-based dielectric materials as discussed below.

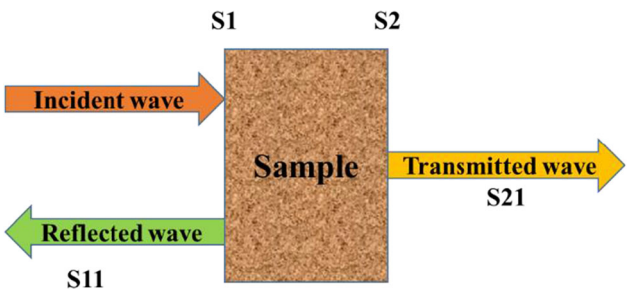


Figure 2 Reflected and transmitted waves with S-parameters.

Transmission/reflection line method

In this measurement technique, the sample is placed inside an enclosed transmission line as shown in Fig. 3. Both the reflected (S_{11}) and transmitted (S_{21}) signals are measured and converted to dielectric parameters using embedded software in VNA using the NRW conversion technique [39].

The advantage of this technique is that it can be used to determine both the permittivity and permeability of the samples under test (SUT). The limitation is to compromise with accuracy due to the air gap effect.

Free space measurement technique

This technique is best suitable for thin flat-faced large materials, and it allows the measurements without any physical contact. A basic free space measurement system consists of (1) a vector network analyzer (VNA), (2) a sample holder, and (3) two-horn antennas as shown in Fig. 4. Two-horn antennas are used as transmitter and receiver, respectively, and the agricultural residue-based SUT is placed in between them. The network analyzer will provide S parameter, i.e., S_{11} and S_{21} [41], and convert it into dielectric parameters using NIST iterative conversion technique.

The advantages of FSMT are: (1) measurement can be done over high frequency and (2) non-destructive measurement technique. The restrictions of this technique are the requirement of large and flat SUT and diffraction effects at the edge of the sample.

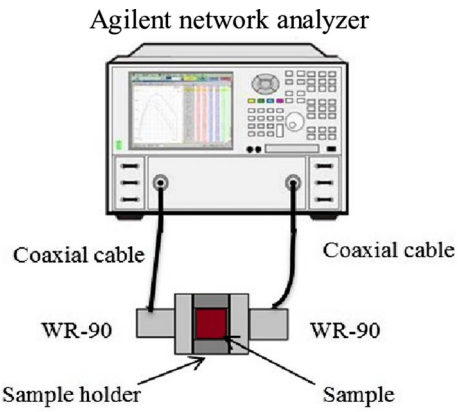


Figure 3: Rectangular waveguide transmission line technique measurement setup [40].

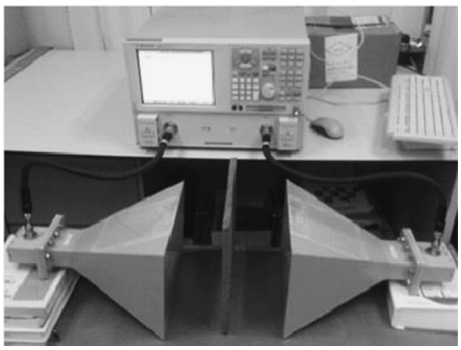


Figure 4 Free space measurement technique [12].

Open-ended coaxial probe(OCP) method

The open-ended coaxial probe (OCP) method is one of the extensively used techniques to extract the electromagnetic properties of liquids and semi-solids materials especially [39]. The dielectric properties of textile materials, powder samples, and other solid samples are also reported [42, 43]. It is also used to measure the reflection parameter (S_{11}) of the samples [1]. It consists of vector network analyzer (VNA), and dielectric assessment kit (DAK) as shown in Fig. 5. DAK consists of a dielectric probe and coaxial cable. The coaxial probe is used to extract the reflection parameters from the sample under test (SUT). It is connected to the VNA that transmits and receives the reflection and transmission signal, respectively. Later on, an inbuilt software in VNA converts the received signal into the S parameter first and then into the form of the phase and amplitude of the reflected signal. Prior to the measure, the probe is standardized using a metallic shorting slab, air, and water [39].

The advantages of this technique are: (1) facile sample preparation, (2) works in a wide range of frequencies, and (3) measurement can be performed in a temperature-controlled environment. The limitations of this technique are: (1) only S_{11} parameter is available, and (2) affected by air gaps.



Figure 5 Experimental setup, **a** dielectric probe, and **b** VNA & DAK.

Sample preparation and development of particle board-based MA

Figure 6 shows the steps for developing an efficient MA structure. At the foremost, the raw samples have been collected and dried under the sun for 3–4 weeks. Afterward, the dry samples have been ground using a ball machine to obtain the uniform particle size in the nanometer dimension. Another method to grind the dry samples which cause non-uniform particle size in the micrometer range is a grinder. The particle size can be measured using an optical microscope as shown in Fig. 7. The absorbers can be fabricated in a different shape as discussed in Sect. 2.5, and the mold has been designed accordingly. The resin and hardener agents have been mixed with the powder form of dried samples, and the mixture has been exposed under ultrasonic wave for 45–60 min to make a uniform mixture. The mixture has been poured inside the mold and kept it until to convert into the hard structure. One can also use hot press machine for compact material forming at high temperature. After that, the dielectric properties of the sample along with S_{11} and/or S_{21} parameters have been measured using the measurement technique as discussed in Sect. 4. The total fabrication procedure is very economical in comparison with conventional microwave absorber. The raw material has minimum cost. The cost for resin, mold, PEC, and electricity to run grinder/ or ball machine are also very minimal. Therefore, the net fabrication cost becomes very less and the fabrication procedure is very simple. On the other hand, the proposed absorber is bio-degradable and eco-friendly which is the prime attribute of agricultural residue-based MA. The total fabrication process is non-toxic in nature under high temperature. To perform a comparative study with simulation results, the obtained values of dielectric parameters are imported in CST Microwave Studio. Thereafter the simulated return loss profile (S_{11}) has been compared with the experimental result.

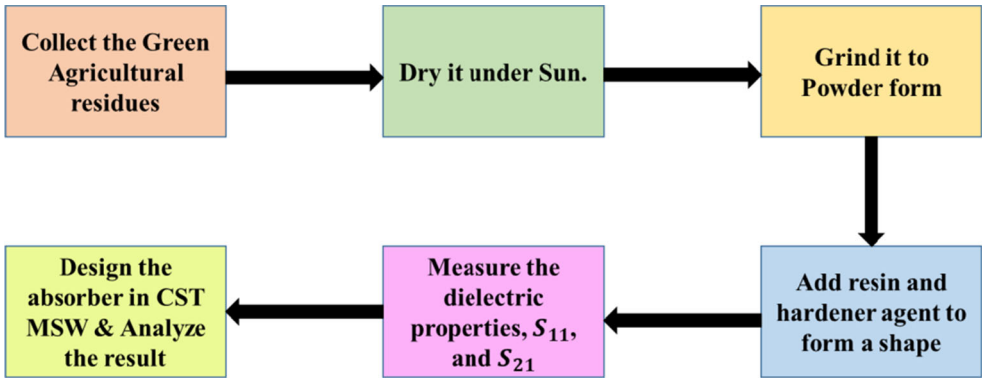


Figure 6 Flow of designing a MA structure using agricultural residue.

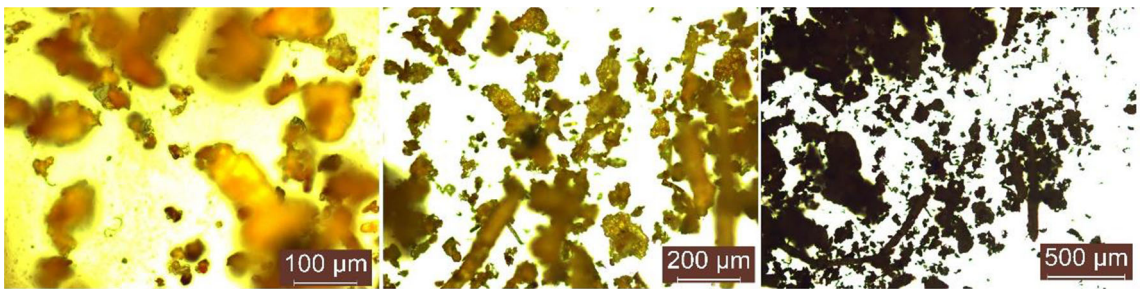


Figure 7 Particle size measurement using an optical microscope (Make: Leica, Model: DM2500 M, Wetzlar, Germany) with 5x optical zoom.

Materials

Agricultural residue-based materials

Agricultural residues with outstanding characteristics such as high permittivity, large carbon content, and low density are a current field of growing interest to develop a low cost, lightweight, and eco-friendly microwave absorber. Agricultural residues such as rice husk, coconut shell charcoal, sugarcane bagasse (SCB), and oil-palm empty fruit bunch have the potential to serve as an alternative microwave-absorbing material [44]. The year-wise number of the published paper on agricultural residue-based MA is shown in Fig. 8.

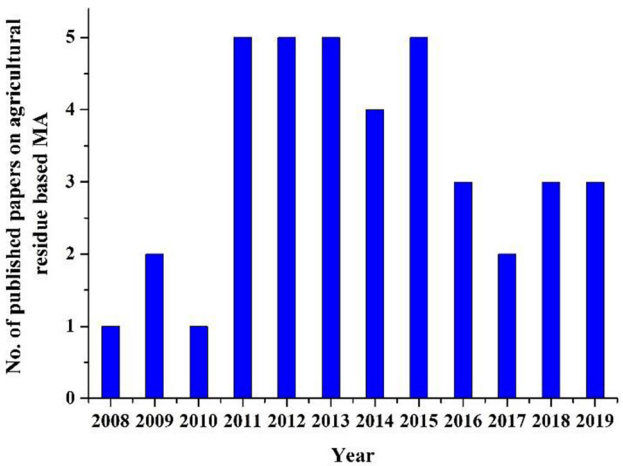


Figure 8 Year-wise number of the published paper.

Rice husk

Rice husk-based pyramidal MA structure was the first time introduced by Nornikman et al. [45]. The free space measurement method was used to measure the dielectric constant, and simulation was performed in CST MWS using the obtained dielectric constant from the free space measurement method. The investigation was studied over the frequency range of 1–20 GHz. Two types of resin, urea formaldehyde (UF) and phenol formaldehyde (PF) are used as a binding and hardener agent. Different percentages of the binding agent such as 10%, 20%, and 30% are added with rice husk. For UF-blended samples, the average return loss was -40.873 dB, -42.121 dB, and -42.312 dB for 10%, 20%, and 30% UF, respectively. For PF-blended samples, the average

return loss was −41.903 dB, −41.779 dB, and −42.309 dB for 10%, 20%, and 30% PF, respectively. The investigation proved that for a particular MAM with different resin mixture would provide different dielectric constant and return loss profile. It is also observed from the investigation that a high value of dielectric constant causes the decrease in wave velocity due to the density of the material. The dielectric properties of the rice husk along with rice straw and kenaf were investigated with different sample distances in the frequency range of 2.2 to 3.3 GHz using free-space measurement technique [46]. Rice husk shows better dielectric properties than other samples. The return loss analysis was reported only for the kenaf sample. Kenaf sample at 10% phenol–formaldehyde exhibits −12.129 dB return loss. Nornikman et al. [47] studied dielectric properties of rice husk particleboard (dimension: 30.3 cm × 30.3 cm × 1.4 cm thickness) using free space measurement technique and microwave absorption performance of pyramidal structure (dimension as per TDK ICT-030 based [48]) at different incidence angles and different resin percentages. The wave incidence angle was 0°, 30°, and 45°. The largest value of dielectric constant 3.68 was achieved using 30% PF content. According to Snell's law, the velocity of the wave at the normal incident is faster than the oblique incident. Faster signal velocities increase the absorption rate for the pyramidal absorber. The best result was obtained at 0° incidences of the microwave in the frequency range of 0.01 to 20 GHz. For UF-blended samples, the average reflection loss obtained in this investigation is −63.1 dB, −63.7 dB, and −69.2 dB for 10%, 20%, and 30% UF, respectively. For PF-blended samples, the average reflection loss obtained is −70.2 dB, −72.9 dB, and −63.5 dB for 10%, 20%, and 30% PF, respectively. According to the study done by Malek et al. [49], the microwave absorption performance can be enhanced by adding rubber tire dust containing high carbon content. Lightweight, elasticity, energy absorption, sound, and heat-insulating properties of rubber tire dust material [50] make the MA structure congenial to provide better MA performance. The rubber tire dust and rice husk were combined in different blending ratios to develop pyramidal MA structure. OCP and FSMT were used to measure the dielectric properties and reflectivity profile of the samples, respectively. The dielectric relaxation parameters obtained in the OCP measurement method were used to design the MA structure in CST MWS simulation software. The maximum average reflection profile of the absorber obtained in the literature is −27.41 dB over the 7–13 GHz frequency range. An 8 × 8 array of pyramidal MA structure using the combination of rice husk and polyester resin was developed in the frequency range of 7–13 GHz [51]. The microwave absorption performance was obtained using the radar cross-section technique. The dielectric properties were measured using FSMT, and later it is used in CST MWS to design the MA structure. The average reflection profile was studied at different points of the array significantly better than −10 dB. The combination of rice husk and rubber tire dust at two different blending ratio-based pyramidal MA structure and its microwave absorption performance was reported in the frequency range of 7–13 GHz [52]. OCP method and FSMT were used to measure the dielectric property and return loss profile of the samples, respectively. Rice husk-rubber tire dust-based (blending ratio: 50% rice husk, 50% rubber tire dust) pyramidal MA exhibited the best performance at the frequency of 11 GHz with −34.96 dB. Iqbal et al. [53] attempted to design rice husk-based geometrically tapered hollow absorber in the anechoic chamber and measured the anechoic performance of the hollow absorbers. Nornikman et al. [54] explained two wedge MA designs using rice husk in the frequency range of 0.01 GHz to 10 GHz. In this study, two designs are proposed where Design A contains two wedges with a height of 15 cm and design B contains a wedge along with the variable height of center from 6 to 14 cm. OCP method was used to calculate the permittivity of the material, and the obtained value was used in CST MWS to design and simulate the mentioned MA structure. Design B of 10 cm center exhibited the best return loss at 9.13 GHz with −77.468 dB. Kaur et al. [55] reported the enhancement in reflection loss by adding coal in rice husk-based MA in the frequency range of 0–20 GHz and also studied a simulative comparative study between rice husk and rice husk-coal composition-based MA. Transmission line technique and radar cross-section (RCS) measurement technique were used to measure the dielectric constant and return loss profile of the proposed MA, respectively. 3 × 3 array pyramidal absorber was designed, and maximum return loss of −52 dB was obtained at 11 GHz. Lee et al. [56] studied the microwave absorption performance of rice husk and ash

composites for different weight ratios in the frequency range from 12.4 GHz to 18 GHz. For dielectric properties measurement, the sample was fabricated with dimensions of 15.799 × 7.899 × 5 mm^3 and the transmission line method was used. 80% rice husk and ash loading exhibited the best dielectric constant of 4.80 and loss tangent of 0.47. The obtained value of dielectric constant and loss tangent was used to simulate the MA structure with dimensions of 300 mm × 300 mm at four different thicknesses such as 5, 10, 15, and 20 mm. The simulation result indicated the maximum absorption at 13.81 GHz frequency with −28.4 dB for 20 mm thickness. Lee et al. [57] introduced a single-layer MA based on rice husk and multi-walled carbon nanotubes (MWCNTs) composites with various weight ratios (3 wt% MWCNTs, 5 wt% MWCNTs, and 15 wt% MWCNTs). Transmission line technique and FSMT were used to measure the dielectric properties and return loss profile of the samples in the frequency range of 8.2–12.4 GHz. Rice husk with 15wt% MWCNTs composition showed the permittivity of 22.82 and loss tangent of 15.45. Rice husk-MWCNTs composition exhibits the maximum absorption performance at the frequency of 9.91 GHz with −33.7 dB. A multilayer MA structure using the composites of rice husk and CNTs was developed in the frequency range of 2–18 GHz by Lee et al. [58]. The flat structure was fabricated with dimensions of 45 cm × 45 cm at two different thicknesses such as 5 and 10 mm. The OCP method was used to measure the dielectric properties of the samples. The obtained value of dielectric properties was used to simulate and optimize the multilayer MA structure in CST MWS. An escalation in return loss < −20 dB was noticed concerning a high lossy composite placed at the bottom layer of multilayer MA. Lee et al. [59] presented rice husk ash and CNTs composites-based single-layer microwave absorber in the frequency range from 2 to 18 GHz. Three different compositions were fabricated with dimensions of 45 cm × 45 cm. OCP method was used to measure the dielectric properties of the samples. The MA structure was designed using the obtained value of dielectric parameters in CST MWS and optimized six different thicknesses such as 1, 2, 3, 4, 5, and 10 mm. The composition exhibited the best return loss point of −20.4 dB at 3.2 GHz at 10 mm thickness. Lee et al. [60] also introduced rice husk-based pyramidal microwave absorber with lossy base layer fabricated using a composition of rice husk and CNTS in the frequency range of 2–18 GHz. OCP method and FSMT were used to measure the dielectric properties and reflection loss profile of the samples. The microwave absorption performance was analyzed at a different angle of incidence from 0° to 60°. Results indicated good microwave absorption characteristics, i.e., −80 dB < RL < −60 dB in the frequency range of 2–6 GHz with a height of 11.5 cm and base layer of 1 cm thickness.

Coconut residues

Salleh et al. [61] reported a single-layer microwave absorber using coconut shell-based activated carbon. Flaxane-80 was used as a binder agent and was mixed in four different ratios (10 wt. %, 15 wt. %, 20 wt. %, and 25 wt. %). The dimensions of the samples were 200 mm × 200 mm × t mm where the thickness t = 3.2, 6.4, 9.6, and 13.8 mm. FSMT was used to measure the complex permittivity as well as the return loss profile of the material. The experimental result exhibited the maximum return loss profile of −22 dB at 9.6 GHz frequency with a 6.4 mm thickness. A study on coco-peat-based microwave absorber in terms of dielectric properties and reflection loss was reported in the frequency range of 8.2–12.4 GHz [62]. The dimensions of the test samples were 22.86 mm × 10.16 mm × 1 mm. OCP method and transmission line technique were used to measure the dielectric properties and return loss profile of the test samples. The result indicated the maximum return loss of −38.914 dB at 10 GHz frequency. An investigation on coconut coir based microwave absorber was reported in terms of the S_{21} parameter in the frequency range of 7.5–12 GHz [63]. The transmission line technique was used to measure the S_{21} parameter of the test samples. This study also revealed the microwave absorption property of other agricultural residues such as corn and lentils. Yah et al. [64] studied the electromagnetic absorption properties of the composite of coconut fiber coir and charcoal powder in terms of complex permittivity and permeability in the frequency range of 8.2–12.4 GHz. Different percentages of charcoal were mixed with the coconut fiber coir to form the composition, and thereafter the epoxy resin with hardener agent was used to bind in the dimensions of 22.866 mm × 10.16 mm. The transmission line technique was used to measure the dielectric properties and return loss

profile of the test samples. The composite with 15 wt. % charcoal content exhibited the best average return loss profile of 29.2444 dB in the entire frequency range.

Sugarcane bagasse (SCB)

SCB-based truncated pyramidal absorber was reported along with its microwave absorption performance in the frequency range of 8.2–12.4 GHz [65]. OCP method was used to measure the dielectric properties of the samples. Therefore, the obtained value of dielectric constant and loss tangent were loaded in CST MWS to simulate and optimize the truncated pyramidal MA structure to get the return loss profile. Three different lengths of the truncated wedge's base were being tested, i.e., 10 cm, 15 cm, and 20 cm. The best return loss point achieved at a 15 cm thickness of −56.07 dB at the frequency of 10.8 GHz. Zahid et al. [66] proposed a pyramidal MA using SCB in the frequency range of 0.1 to 20 GHz. The dielectric constant and loss tangent of SCB were measured using the OCP method and then used to design the MA structure in CST MWS. The dimensions of the base were 15 cm × 15 cm × t cm (t = 1, 2, 3, and 5 cm), and the pyramidal body was 5 cm × 15 cm × 13 cm ($W \times L \times H$), respectively. The best reflection loss is −85.498 dB at 11.58 GHz frequency with 1 cm base thickness.

Rubber wood sawdust

Activated carbon prepared from rubberwood sawdust by chemically activated and polyurethane composites was reported for microwave absorption in the frequency range of 1–3 GHz [67]. A sample of 4.8 cm thickness exhibited the best reflection loss of −10.4 dB in the frequency range of 1.8–1.9 GHz.

Banana leaves

3 × 3 array pyramidal MA structure using dried banana leaves was found to be a potential alternative microwave absorber [68]. Polyester resin along with methyl ethyl ketone peroxide (MEKP) was used to fabricate the particleboard. The dimensions of the particleboard were: W = 15 cm, L = 15 cm, and T = 2 cm for base part and W = 5 cm, L = 15 cm, and H = 13 cm for the pyramidal part where W, L, T, and H were the width, length, thickness, and height, respectively. The OCP method was used to measure the dielectric properties of the particleboard. The dielectric properties were used to design and optimize the pyramidal MA structure in CST MWS in the frequency range from 0.01 GHz to 20 GHz. The result showed the best reflection loss of −123.47 dB at 4.0 GHz. The improvement in reflection loss of 3 × 3 array pyramidal MA using the composition of dried banana leaves and coal was reported over 8.2–12.4 GHz frequency range [69]. The pyramidal height and base dimensions were 13 cm and 15 cm × 15 cm × 2 cm, respectively. Transmission line technique and FSMT were used to measure the dielectric parameters and microwave absorption performance, respectively. The experimental result indicated that the best return loss point is at 12 GHz with −49 dB. The MA structure was further designed and optimized in CST MWS in the frequency range of 0–20 GHz. The simulative result depicted that the maximum value of reflection is −76 dB at 16 GHz. The microwave absorption performance of dried banana leaves based single-layer MA was studied in terms of dielectric properties in the frequency range of 1–20 GHz [1]. The dimensions of the samples are given in Table 2. The dielectric properties were measured by the open-ended coaxial probe method as shown in Fig. 9 and Table 3.

The reflectivity parameter (S_{11} dB) was measured using an open-ended coaxial probe method as shown in Fig. 10. The MA structure was also designed in CST Microwave Studio (CST MWS) using the obtained experimental value of dielectric constant and loss tangent and the result also depicted in Fig. 10. Both the results validate that sample with a thickness of 16 mm poses the best return loss point at 1.133 GHz as shown in Table 4.

The microwave absorption properties along with electromagnetic parameters of the double-layer MA using dried banana leaves were studied in the frequency range of 1–20 GHz [70]. The dimensions of the samples are given in Table 5. The dielectric properties of the samples were measured using an open-ended coaxial probe method as shown in Fig. 11 and Table 6.

A comparative study of return loss profile (S_{11} dB) between CST MWS simulative result and the experimental result is illustrated in Fig. 12 and Table 7. Both the result validate that the sample with the double-layer MA structure exhibits the best return loss value of −32.61 dB at 8.341 GHz.

Table 2 Sample preparation and dimensions of single-layer MA using banana leaves [1]

Samples	Percentage of resin and hardener agent (wt. %)	Mould dimensions	Sample thickness (measured) (mm)	Sample thickness with PEC (mm)
A_1	33.3	5 cm × 3 cm × 8.16 mm	8.16	8.37
A_2	33.3	5 cm × 3 cm × 15.98 mm	15.98	16.19
A_3	33.3	5 cm × 3 cm × 32.02 mm	32.02	32.23

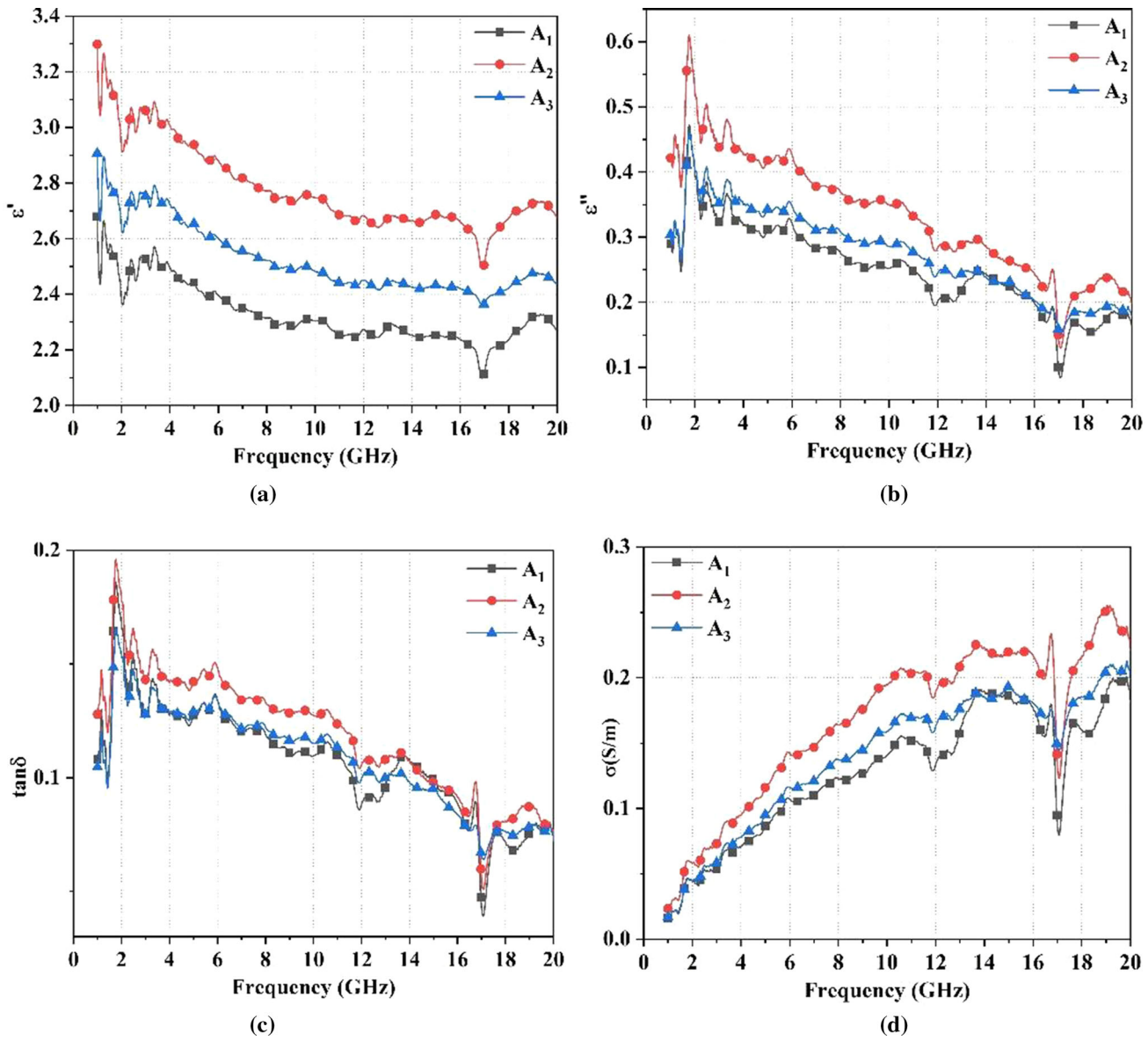


Figure 9 Electromagnetic parameters of single-layer absorber for 8 mm, 16 mm, and 32 mm thickness, respectively, **a** dielectric constant, **b** loss factor, **c** loss tangent, and **d** conductivity [1].

Table 3 Average dielectric properties of the single-layer absorber using dried banana leaves [1]

Samples	ε′	ε″	tanδ	σ (S/m)
A_1	2.33	0.25	0.107	0.13
A_2	2.78	0.33	0.119	0.17
A_3	2.52	0.28	0.108	0.14

Resins and hardener agents

Resin with hardening agents is used to bind the particles of the samples and the consecutive two layers which transmits strength by diminishing the air gap between the layers of constituents and therefore protect it from abrasion. Different type's resins such as urea–formaldehyde (UF), phenol–

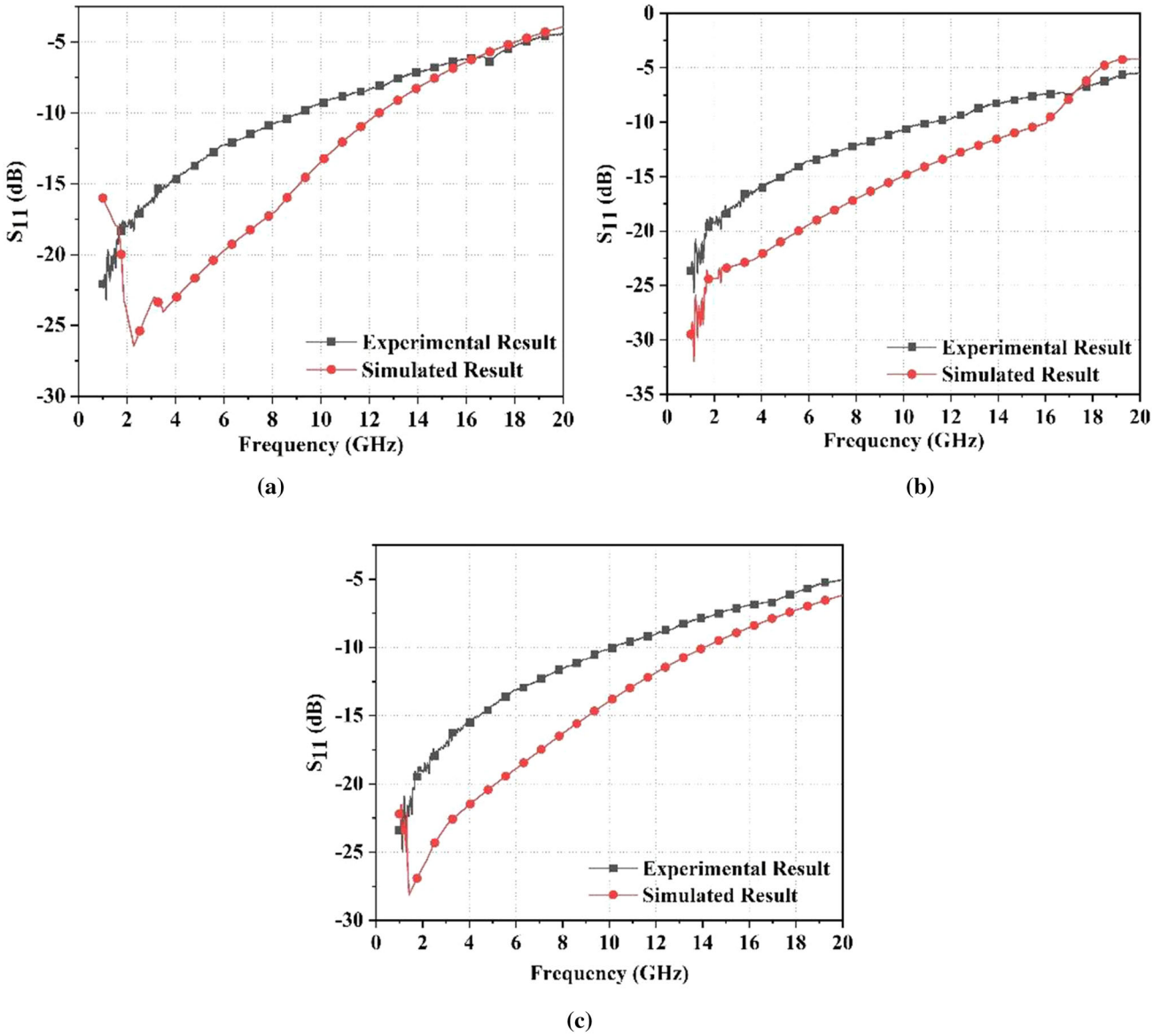


Figure 10 Comparative validation of reflection loss of single-layer MA for the samples at 8 mm, 16 mm, and 32 mm thickness, respectively: **a** A1, **b** A2, and **c** A3 [1].

Table 4 The maximum value of the absorption rate of the single-layer absorber for various thicknesses at the resonant frequency [1]

Samples	Experimental result			Simulated result		
	S_{11} (dB)	Absorption rate (%)	RL < −10 dB at freq. (GHz)	S_{11} (dB)	Absorption rate (%)	RL < −10 dB at freq. (GHz)
A_1	−23.187	99.52	1.133	−26.444	99.77	2.254
A_2	−25.663	99.73	1.133	−31.991	99.94	1.133
A_3	−24.974	99.68	1.133	−28.131	99.85	1.418

Table 5 Sample preparation and dimensions of double-layer MA using banana leaves [70]

Samples	Materials	Dimension (cm^2)	Thickness (mm)
Sample_1	Single-layer particle board Banana leaves powder: 4 gm Binding agent: 16 ml	6 × 6	4
Sample_2	Single-layer particle board Banana leaves powder: 4 gm Charcoal (purity: 99.9%, dimension: 60–70 μm): 30 mg Binding agent: 16 ml	6 × 6	4
Sample_3	Double layer particle board with PEC Sample_1 Sample_2 Binding agent: 16 ml Thickness of PEC: 0.02 mm	6 × 6	8.02

formaldehyde (PF), Flaxane-80, epoxy resin, and polyester resin were used in the previous literature. Currently, the epoxy resin and polyester resin are used widely and the properties, as well as comparative studies of these resins, are given in Table 8.

Simulation strategy

To analyze the microwave absorption performance under given boundary condition, CST Microwave Studio (CST MWS) 2019 3D Electromagnetic Simulation Software for high-frequency components is used as a simulative tool. Due to dispersive dielectric properties, the MA can be developed and simulated in CST MWS using the value of dielectric constant and imaginary part of the complex permittivity or loss tangent from the experimental measurement. The finite difference time domain numerical technique based on transient solver is used to enact three-dimensional (3D) simulations with their transverse EM mode boundary conditions. To emulate the free space reflectivity evaluation setup, waveguide port is used to excite the signal in the far-field region with the open boundary condition of x-axis set as E-field = 0; the y-axis set as H-field = 0, and the positive z-axis set as the path of EM wave propagation toward the negative z-axis direction [1].

Future prospects of agricultural residue-based MAMs

Microwave absorber using agricultural residue-based MAMs is explicitly discussed systematically. Apart from microwave absorber, agricultural residue-based MAMs can be used in several applications such as electromagnetic compatibility, antenna applications, microwave devices, and various microwave measurements as shown in Fig. 13. The ultra-wideband (UWB) microstrip patch antenna on agricultural residue-based substrate was designed and developed by Ullah et al. [73]. The agricultural residue substrate-based designed antenna exerted good radiation performance over the UWB spectrum. The anechoic chamber can be developed for microwave

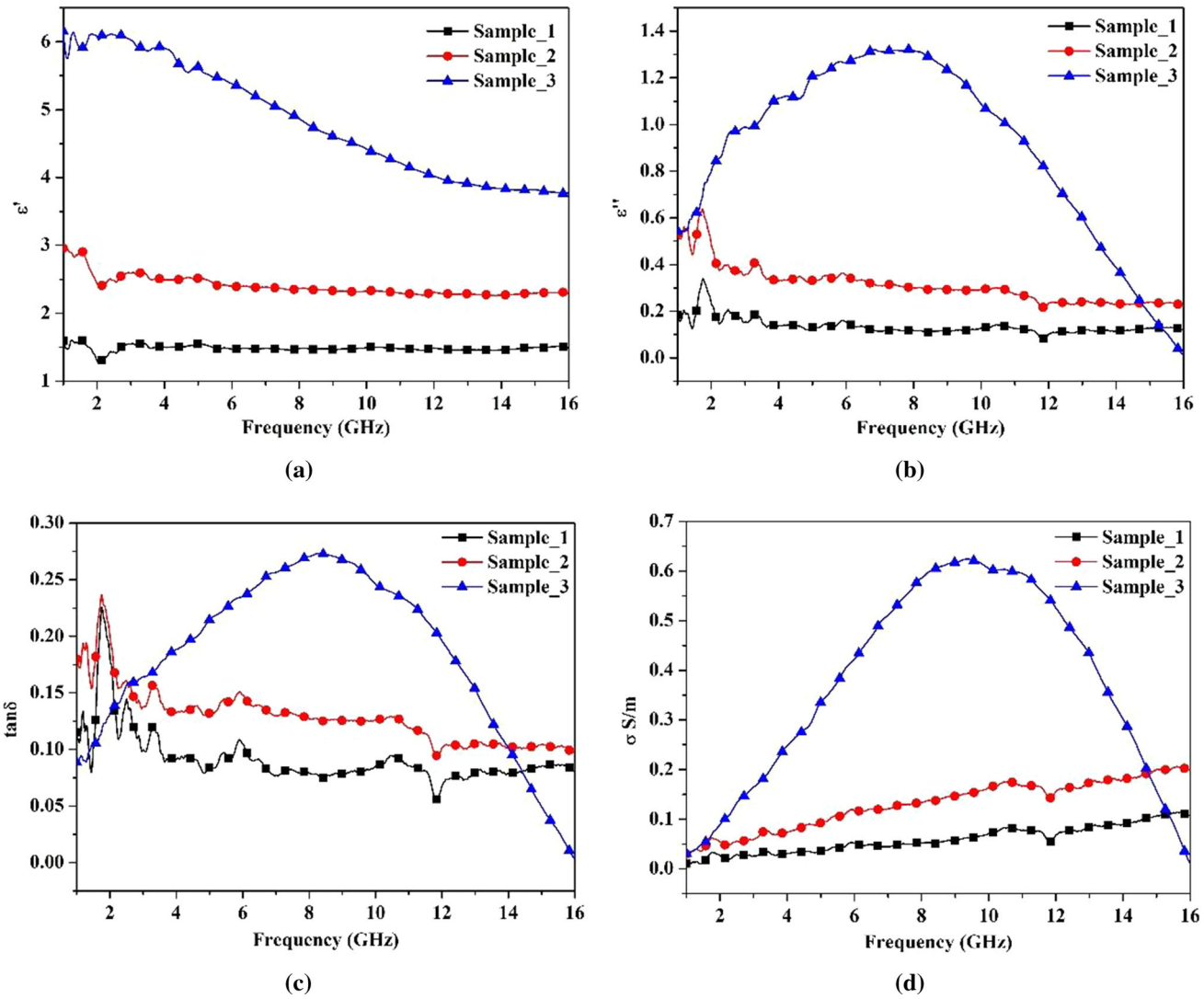


Figure 11 Electromagnetic parameters of the samples: **a** dielectric constant, **b** loss factor, **c** loss tangent, and **d** conductivity [70].

Table 6 Average dielectric properties of the samples [70]

Samples	Dielectric constant (ε′)	Loss factor (ε″)	Loss tangent (tan δ)	Conductivity (σ)
Sample_1	1.48	0.135	0.091	0.075
Sample_2	2.37	0.293	0.122	0.15
Sample_3	4.64	0.711	0.15	0.30

measurements like antenna radiation patterns, electromagnetic interference, and a study on anechoic performance was studied by Iqbal et al. [53]. Therefore, agricultural residue-based MAMs can be used to fabricate low cost and lightweight anechoic chamber. A compact two-element multiple-input multiple-output (MIMO) antenna proposed for triple-band applications with very high port isolation is designed on coconut husk-based substrate [74]. The dielectric properties of the substrate are depicted in Fig. 14. Frequency bands of the antenna for ($S_{11} \leq -10$ dB) are obtained from 2.04–2.51 GHz, 4.43–5.35 GHz and 6.76–8.78 GHz. The antenna fulfills the WLAN (2.4–2.485/5.15–5.35 GHz), fixed mobile (4.45–5.15 GHz) and ITU (8–8.5 GHz) application bands [74].

Corn syrup was found as a promising material to mimic the dielectric properties of breast tissues and

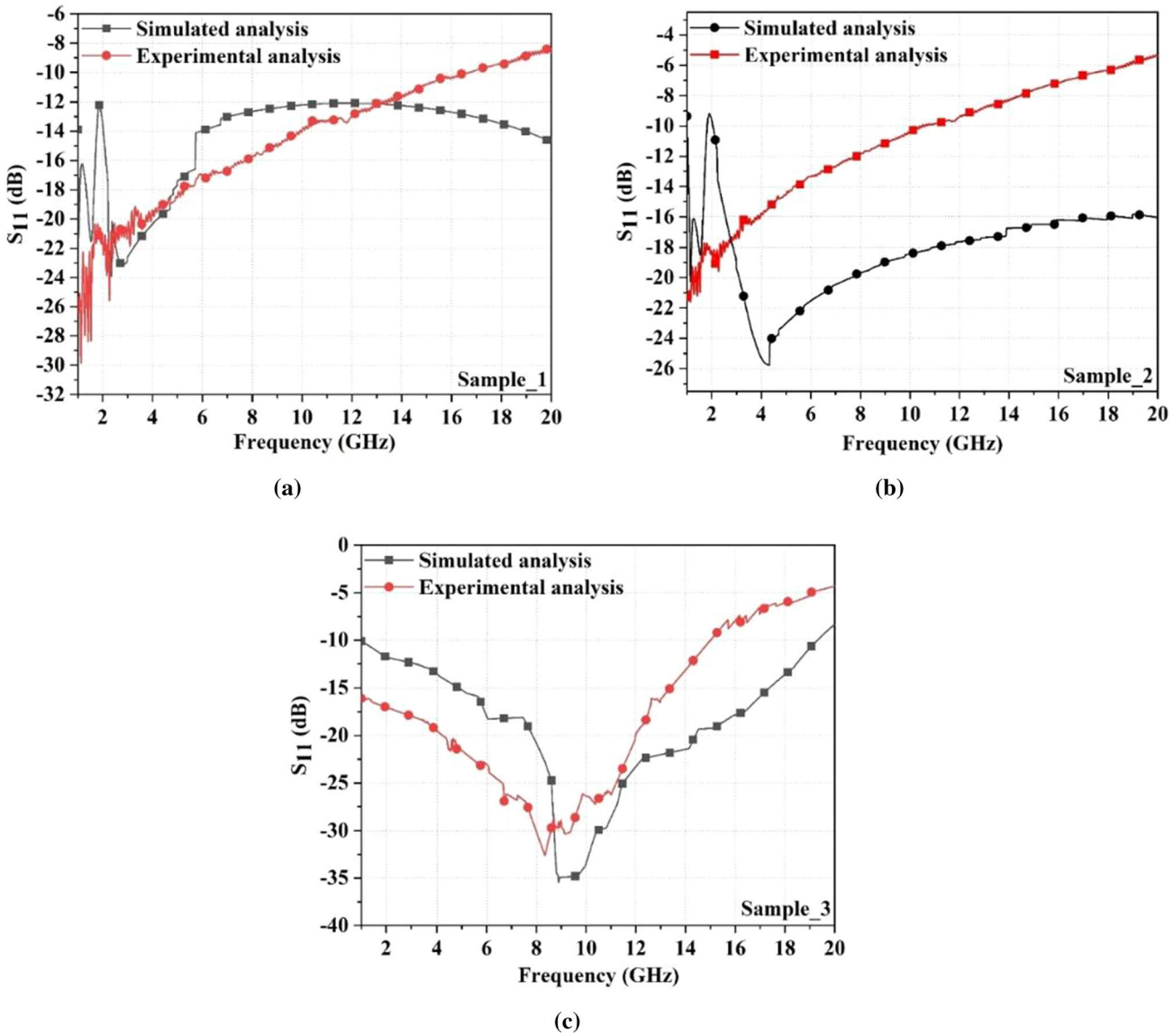


Figure 12 Comparative study on return loss profile obtained from simulated analysis and experimental analysis of three samples: **a** Sample_1, **b** Sample_2, and **c** Sample_3 [70].

Table 7 Maximum return loss profile of the samples [70]

Samples	Measured value		Simulated value	
	S_{11} (dB)	RL < −10 dB freq. (GHz)	S_{11} (dB)	RL < −10 dB freq. (GHz)
Sample_1	−21.63	1.133	−23.91	4.629
Sample_2	−29.84	1.133	−25.76	2.577
Sample_3	−32.61	8.341	−35.41	8.752

can be used as an effective coupling medium in microwave imaging to reduce scattering which in turn enhances resolution [75]. Jung et al. [76] proposed cellulose nanofibril extracted from wood-based potential microwave substrate for different RF and digital electronic devices. The agricultural residue-based MA shows promising absorption performance as depicted in Table 9. Uniform particle size in nanometer can enhance the microwave absorption performance of agricultural residue-based MAs.

Table 8 Resins and their properties

Resin	Properties
Epoxy Resin	Excellent chemical resistance, particularly to alkaline environments
Hardener agent: epoxy	Outstanding adhesion to a variety of substrates
	Very high tensile, compressive, and flexural strengths
	Excellent electrical insulation properties and retention thereof on aging or exposure to difficult environments
	Remarkable resistance to corrosion
Polyester resin	Adequate resistance to water and a variety of chemicals
Hardener agent: methyl ethyl ketone peroxide (MEKP)	Adequate resistance to weathering and aging
	Low cost
Accelerator: cobalt	Polyesters can withstand a temperature up to 80 °C
	Polyesters have good wetting to glass fibers

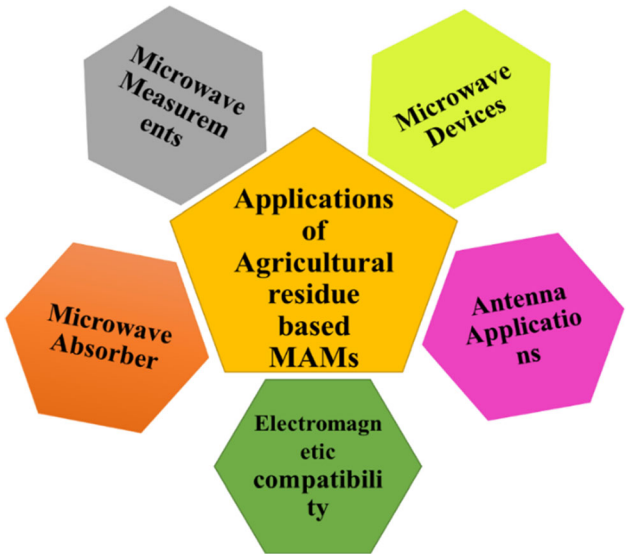


Figure 13 Various applications of agricultural residue-based MAMs.

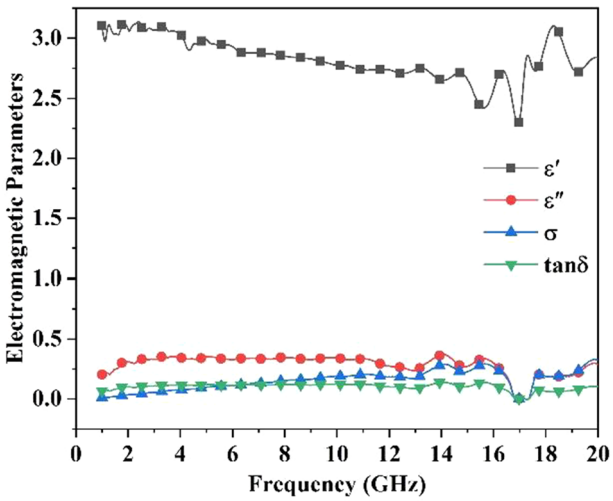


Figure 14 Dielectric properties of coconut husk-based substrate for antenna design.

Further, the absorption performance can be increased using a multilayer structure which is under investigation. There is a scope of further improvement in absorption performance by decreasing the thickness and increasing the absorption bandwidth for usage in different applications such as military applications, and communication sectors. Eventually, the cheap fabrication, eco-friendly and lightweight characteristics make the agricultural residue-based MA an inevitable and potential alternative to the world in the coming decade.

Conclusions

This review attempts to uphold an empirical vignette on various types of agricultural residue-based microwave-absorbing material. An extensive discussion is made on different microwave absorption mechanisms, loss mechanism, polarization techniques, and the related factors affecting said mechanism. Also, different types of microwave absorbers, measurement techniques, and their utility are explained in detail. Various sample fabrication and design techniques are mentioned systematically to help scientists and engineers for the fabrication of microwave-based MA. Finally, we tried to uphold a futuristic glimpse of agricultural residue-based microwave-absorbing material which can be a

Table 9 Agricultural residue-based microwave absorber

Agricultural residues	Dimensions	Shape	RL min. (dB)	RL < −10 dB Frequency (GHz)	Ref
Rice husk	As per TDK ICT-030	Pyramidal	− 85.947	7.98	[45]
Rice husk + rubber tire dust	As per TDK ICT-030	Pyramidal	− 27.41	7–13	[51]
Rice husk + rubber tire dust	As per TDK ICT-030	Pyramidal	− 34.96	11	[52]
Rice husk	H_{w1} = 15 cm, H_{w2} = 10 cm, H_b = 2 cm, W_b = 15 cm, L_b = 10 cm	Wedge	− 77.468	9.13	[54]
Rice husk + coal	2 cm thick base	3 × 3 array pyramidal	− 52	11	[55]
Rice husk + ash	300 mm × 300 mm × 20 mm	Single layer	− 28.4	13.81	[56]
Rice husk + MWCNTs	300 mm × 300 mm × 8 mm	Single layer	− 33.7	9.91	[57]
Rice husk + CNTs	450 mm × 450 mm × 10 mm	Multilayer			[58]
Rice husk ash + CNTs	450 mm × 450 mm × 10 mm	Single layer	− 20.4	3.2	[59]
Rice husk	Height = 11.515 cm, base layer thickness = 1 cm	9 × 9 array pyramidal	− 80 < RL < −60	2–18	[60]
Coconut shell	200 mm × 200 mm × 6.4 mm	Single layer	− 22	9.6	[61]
Coco-peat	22.860 mm × 10.16 mm × 1 mm	Rectangular mold	− 38.914	8–12	[62]
Coconut fiber coir + charcoal powder	22.860 mm × 10.16 mm	Rectangular mold	− 29.2444	8.2–12.4	[64]
Sugarcane bagasse	Thickness of the base layer = 15 cm	Truncated wedge	− 56.07	10.8	[65]
Sugarcane bagasse	Base part: W = 15 cm, L = 15 cm, and T = 2 cm, pyramidal part, W = 5 cm, L = 15 cm, and H = 13 cm	Pyramidal	− 85.498	11.58	[66]
Rubber wood sawdust	300 mm × 300 mm × 48 mm	Flat shaped	− 10.4	1.8–1.9	[67]
Banana leaves	$W \times L \times T$ = 15 × 15 × 2(cm^3) Pyramidal height = 13 cm	Pyramidal	− 49	12	[69]
Banana leaves + charcoal			− 76	16	
Banana leaves	5 cm × 3 cm × 15.98 mm	Single layer	− 25.663	1.133	[1]
Banana leaves	6 cm × 6 cm × 8.02 mm	Double layer	− 32.61	8.341	[70]
Rice husk + CNT	200 mm × 200 mm × 2 mm	Single layer	− 18.5	14.4	[71]
Rice husk	W = 15 cm, L = 15 cm,	Pyramidal	− 46.55	7.6	[72]
Sugarcane bagasse			− 38.33	8.2	
Banana leaves			− 43.13	6.6	
Rice husk	$W \times L \times H$ = 15 × 15 × 15(cm^3)	Pyramidal	− 38.223	0.01 ≤ RL ≤ 20	[68]
Banana leaves			− 82.302		

potential alternative to serve as an eco-friendly microwave absorber.

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